

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

Sravya(192472192)

I Narappagari (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Sravya

16
20

B

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart home automation System is being designed to manage lightening, temperature, and security based on user behaviour & environmental conditions.

1) ~~Simple Goal test~~ goal based agent
Based on the output it searches the option available to the goal accurately. it chooses best action with low risk to get goal reached.

- the System get inputs from the sensors like temperature, etc.
- According to the temperature it adjust the temperature in AC and fans
- when motion is detected the lights will turn on & if motion is not detected the lights will turn off.
- robot vaccum cleaner is an example of online search algorithm in which it does not know about the algorithm (room's layout) by working on the system it learns. first it moves if the dust is detected it will collect it & any object or person is detected it will

Change the direction

- with the help of IoT Sensors we can control the device through our mobile.
- The system will detect & respond real time inputs as motion detection, temperature change & time of day

Simple reflex agent

- It will get the input from the sensor to work. It won't store any previous inputs.

2. Uniform Cost Search:

It calculates the total path a agent takes to reach goal state from initial state.

for example:

A student needs to go to college from home. he has 3 routes to go to college

A $\rightarrow$ 5 km

B $\rightarrow$ 4 km

C $\rightarrow$ 4.5 km

It chooses path B because it is

Having less distance also takes less time.

- As a robot which is placed in a unknown maze find a path to the exit without prior knowledge of the layout.
 - for example robot vacuum cleaner it doesn't know about environment still work by using Camera it detects the objects, persons, dust. it collect dust & waste if objects or persons it will move in another direction.
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